

FOR LEARNERS - KEEP FOR ALL 12 MODULES



Glossary of Terms

Enterprise Data Engineering with SQL, Redis & Kafka

120 Hours | 12 Modules

How to use this glossary

This programme introduces roughly two hundred technical terms across 120 hours. Nobody remembers them on first hearing, and you are not expected to.

Keep this sheet beside you for all twelve modules. Every entry gives the term, the module where you first meet it, and a plain definition written for someone meeting it for the first time. Where a banking example makes the idea concrete, it is included.

Terms are alphabetical, not grouped by module, because you will look things up in the middle of a lab rather than in the middle of a lecture.

A note on the definitions. These are working definitions, deliberately short. The course material is where the full explanation lives. If a definition here leaves you unsatisfied, that is the signal to go back to the module.

A

Term	Module	What it means
ACID	1	The four guarantees a transaction gives: atomicity, consistency, isolation and durability. The reason a bank uses a database rather than files.
<code>acks</code>	8	A Kafka producer setting controlling how many brokers must confirm a write. A bank uses <code>acks=all</code> .
Aggregate function	3	A function that turns many rows into one value, such as <code>SUM</code> , <code>COUNT</code> or <code>AVG</code> .
Alias	2	A temporary name given to a column or table in a query, using <code>AS</code> .
ANSI SQL	2	The standard every database partly follows and partly extends. Why syntax differs between Oracle and PostgreSQL.
Anti-join	3	A join pattern that returns rows from one table with no match in another. Used to find accounts with no card.
Anomaly (insert, update, delete)	4	A problem caused by storing the same fact in many places. Fixed by normalization.
Archive log	9	A saved copy of a write-ahead log file, needed for point-in-time recovery.
Atomicity	1	All statements in a transaction happen, or none do. A transfer never debits without crediting.
At-least-once	8	Delivery where a message is never lost but may arrive twice. What a bank chooses, paired with idempotent consumers.
At-most-once	8	Delivery where a message never arrives twice but may be lost. Unacceptable for money.
Autocommit	6	The default mode where every statement commits on its own. The reason an unguarded two-statement transfer can half-complete.

B

Term	Module	What it means
Base backup	9	A full physical copy of the database files, the starting point for point-in-time recovery.
<code>bcp</code>	10	A SQL Server command-line tool for bulk copying data in and out of a table.
BCNF	4	Boyce-Codd normal form. A stricter 3NF: every determinant must be a candidate key.
Big-bang migration	11	Moving everything in one outage window. Simple to plan, expensive to get wrong.
Broker	8	One Kafka server. A cluster is several brokers holding partitions between them.
B-tree	5	The index structure almost every database uses. Balanced, so finding one row in 40 million costs about four page reads.
Buffer cache	1	Memory holding recently used data pages so the engine avoids reading disk.
<code>BULK INSERT</code>	10	The SQL Server statement that loads a delimited file into a table quickly.

C

Term	Module	What it means
Cache-aside	7	The caching pattern where the application checks the cache, and on a miss reads the database and stores the result.
Cache hit rate	7	The share of reads answered by the cache. The number that says whether caching is working.
Cache stampede	7	Many requests missing the same expired key at once and all hitting the database together.
Candidate key	4	Any column or set of columns that uniquely identifies a row. The primary key is the one you chose.
CAP theorem	7	Under a network partition a distributed system must choose between consistency and availability.
Cardinality	1	The number of rows in a table. Also, in an ER diagram, how many of one thing relate to another.
CDC	8, 11	Change data capture. Reading a database's own log to learn what changed, rather than polling.
CHECK constraint	2	A rule the database enforces on a column's values, such as <code>balance >= 0</code> .
Checkpoint	1	The moment the engine flushes dirty pages to disk, bounding how much log must be replayed after a crash.
Clustered index	10	In SQL Server, the index that is the table. A table has at most one.
Composite index	5	An index over more than one column. Column order decides which queries it serves.
Consumer group	8	A set of Kafka consumers sharing the work of a topic. Each partition goes to exactly one member.
Consumer lag	8	How far behind a consumer group is, in messages. The first metric to alert on.
Correlated subquery	3	A subquery that refers to the outer query and is evaluated once per outer row.
Covering index	5	An index holding every column a query needs, so the table itself is never read.
Crow's foot	4	The ER diagram notation where the many end of a relationship is drawn as a three-pronged fork.
CTE	6	Common table expression. A named subquery defined with <code>WITH</code> , readable and reusable within one statement.
Cursor	5	A row-at-a-time loop over a result set. Almost always slower than the set-based alternative.

D

Term	Module	What it means
DACPAC	11	A SQL Server package holding schema only. The one that belongs in source control.
Data Pump	11	Oracle's export and import utility, <code>expdp</code> and <code>impdp</code> .
Data independence	1	The ability to change storage or schema without rewriting applications. The reason <code>SELECT *</code> is discouraged.
DCL	2	Data control language. <code>GRANT</code> and <code>REVOKE</code> .
DDL	2	Data definition language. <code>CREATE</code> , <code>ALTER</code> , <code>DROP</code> . Changes structure.
Deadlock	6	Two transactions each holding a lock the other needs. The engine kills one. Prevented by consistent lock ordering.
<code>DELETE</code>	2	Removes rows, can be filtered with <code>WHERE</code> , can be rolled back, fires triggers.
Denormalization	4	Deliberately storing a derived or repeated value for speed, accepting the update cost. <code>account.balance</code> is one.
Dirty read	6	Reading data another transaction has written but not committed.
DML	2	Data manipulation language. <code>INSERT</code> , <code>UPDATE</code> , <code>DELETE</code> , <code>SELECT</code> . Changes data.
DMV	10	Dynamic management view. SQL Server's built-in views exposing what the engine is doing.
Dual write	8	Writing to the database and to Kafka as two separate operations. The bug the outbox exists to prevent.
Durability	1	Once committed, a change survives a crash. Delivered by the write-ahead log.

E

Term	Module	What it means
Eviction policy	7	What Redis discards when it reaches <code>maxmemory</code> . <code>allkeys-lru</code> for a cache.
<code>EXCEPT</code>	3	A set operator returning rows in the first query but not the second. The core of migration reconciliation.
Exactly-once	8	Delivery where a message is processed once and only once. In practice, at-least-once plus idempotent consumers.
Execution plan	5	The engine's chosen strategy for a query. Read it before tuning anything.
<code>EXPLAIN ANALYZE</code>	5	The PostgreSQL command that runs a query and shows the plan with real timings and row counts.
Extent	10	Eight contiguous SQL Server pages, 64 KB. The unit of space allocation.

F

Term	Module	What it means
Fan-out	3	The row multiplication that happens when you join to a one-to-many table. The reason a COUNT can silently triple.
Fill factor	10	How full an index page is left when built, trading space for fewer page splits.
FILEGROWTH	10	How much a SQL Server file grows when full. Set it in megabytes, never as a percentage.
FLASHBACK_TIME	11	The Data Pump option giving a consistent point-in-time export across all tables.
Foreign key	1	A column pointing at another table's primary key. The database refuses to break the link.
FOR UPDATE	6	Locks the rows a SELECT returns until the transaction ends. The standard fix for a lost update.
Frame (window)	6	The set of rows a window function looks at for the current row. ROWS counts rows, RANGE counts values.
Functional dependency	4	When one column's value determines another's. The basis of every normal form.

H

Term	Module	What it means
HAVING	3	Filters groups after aggregation, where WHERE filters rows before it.
Heap	10	A SQL Server table with no clustered index. Rows sit in no particular order.
Heterogeneous migration	11	Moving between different database products, such as Oracle to PostgreSQL. Schema, code and types all change.
Homogeneous migration	11	Moving between instances of the same product. Far simpler.
Hot standby	9	A replica that accepts read-only queries while it replays changes from the primary.

I

Term	Module	What it means
Idempotent	8	An operation that can be repeated with no additional effect. What makes at-least-once delivery safe.
Index-only scan	5	A read satisfied entirely from the index. Heap Fetches: 0 in the plan confirms it.
Indexed view	10	A SQL Server view with its results physically stored and maintained. Comparable to a materialized view that never goes stale.
inserted and deleted	10	The two virtual tables inside a SQL Server trigger holding new and old rows.
Invalidation	7	Removing a cached value when the underlying data changes. Delete it, and do it after the commit.
IFSC	1	The eleven-character code identifying an Indian bank branch, such as ICIC0000101 .
Isolation level	6	How much concurrent transactions can see of each other. Read committed, repeatable read, serializable.

J

Term	Module	What it means
Join	3	Combining rows from two tables on a matching condition.
Junction table	4	The table that resolves a many-to-many relationship, holding a foreign key to each side.

K

Term	Module	What it means
Kafka	8	A distributed, durable, replayable log of events. Not a queue: reading does not remove.
Key lookup	10	The extra read SQL Server does when a non-clustered index lacks a needed column. Removed by an INCLUDE .
Keyset pagination	5	Paging with WHERE id < last_seen instead of OFFSET . Stays fast at page 10,000.
KRaft	8	Kafka's own consensus mode, replacing the old ZooKeeper dependency.

L

Term	Module	What it means
Latency	7	How long one operation takes, as experienced by the caller.

Term	Module	What it means
<code>LIMIT</code>	2	Caps the number of rows returned. <code>TOP</code> in SQL Server.
Lock	6	A reservation on a row or table that makes other transactions wait.
<code>lock_timeout</code>	6	How long a statement waits for a lock before giving up. Better than waiting forever.
Logical backup	9	A backup made of SQL statements, such as <code>pg_dump</code> . Portable, slow to restore, restores single tables.
Logical replication	11	Replication that ships row changes rather than disk blocks, so source and target can differ.
Lost update	6	Two transactions read the same value, both write, and one change disappears.
LRU	7	Least recently used. The eviction rule that discards whatever has gone untouched longest.

M

Term	Module	What it means
Materialized view	5	A view whose results are stored on disk and refreshed on command. Fast to read, potentially stale.
<code>maxmemory</code>	7	The memory ceiling for Redis. Leaving it unset is how a cache takes down a server.
<code>min.insync.replicas</code>	8	How many Kafka replicas must be in sync for an <code>acks=all</code> write to succeed. Must be at least 2 to mean anything.
MVCC	6	Multi-version concurrency control. Each transaction sees a consistent snapshot, so readers never block writers.

N

Term	Module	What it means
N+1 query	9	Running one query, then one more per returned row. 200 queries where 2 would do.
Non-clustered index	10	A separate index structure pointing back at the table. A SQL Server table can have many.
Normalization	4	Organising columns so each fact is stored exactly once. 1NF through BCNF.
<code>NOT EXISTS</code>	3	The safe way to ask "has no matching row". Unlike <code>NOT IN</code> , it handles NULLs correctly.
<code>NOT IN</code> with NULL	3	Returns no rows at all if the subquery contains a single NULL. A silent, long-lived reporting bug.
NoSQL	7	Databases that are not relational: key-value, document, column-family and graph.
NULL	2	Unknown, not zero and not empty. Any comparison with NULL is unknown, which is why <code>IS NULL</code> exists.

O

Term	Module	What it means
Offset	8	A message's position in a Kafka partition. A consumer group commits how far it has read.
OLAP	1	Analytical workload. Few queries, each reading millions of rows. The reporting warehouse.
OLTP	1	Transactional workload. Many small reads and writes. Core banking.
Optimistic locking	6	Not locking, but checking on write that nothing changed. Suits screens where a human thinks between read and write.
ora2pg	11	The tool that assesses and converts an Oracle schema to PostgreSQL. Its cost estimate is an input, not an answer.
Orphaned user	10	A SQL Server database user whose matching server login is missing, usually after a restore onto a different server.
Outbox	8	A table written in the same transaction as the business change, from which a relay publishes events. Solves dual write.

P

Term	Module	What it means
PAN	9	In Indian banking, the Permanent Account Number, a tax identifier. Sensitive data requiring encryption or masking.
Parameter sniffing	10	SQL Server caching a plan built for the first parameter it saw, which suits that value and no other.
Partition (Kafka)	8	One ordered log within a topic. Ordering is guaranteed inside a partition, never across them.
Partition key	8	The value deciding which partition an event lands in. Key by <code>account_id</code> so one account's events stay ordered.
Partial index	5	An index with a <code>WHERE</code> clause, covering only the rows you query. Small and cheap.
pgcrypto	9	The PostgreSQL extension providing column-level encryption functions.
Physical backup	9	A byte-level copy of the data files. Fast to restore, all-or-nothing, required for point-in-time recovery.
PITR	9	Point-in-time recovery. Restoring a base backup and replaying the log to a chosen instant.
Plan cache	10	Where the engine keeps compiled execution plans for reuse.
Primary key	1	The column identifying a row uniquely. Never NULL, never reused.
Pub/Sub	7	Redis publish and subscribe. Fire and forget: a subscriber that is not listening never gets the message.

Q

Term	Module	What it means
Query optimizer	5	The component that chooses how to execute a query, using statistics about your data.

R

Term	Module	What it means
RAISERROR	10	The older SQL Server error-raising statement. Prefer THROW.
Reconciliation	11	Proving source and target hold the same data. Three levels: counts, aggregates, row by row.
Recovery model	10	SQL Server's setting controlling log retention. Full, bulk-logged or simple. Full without log backups fills the disk.
Recursive CTE	6	A CTE that refers to itself, used for hierarchies such as an org chart. Always add a cycle guard.
Redis	7	An in-memory key-value store used here as a cache, a session store and a rate limiter.
Referential integrity	1	The guarantee that a foreign key always points at a row that exists.
Relay	8	The process that reads unpublished outbox rows and publishes them to Kafka.
Replication slot	11	A PostgreSQL object holding WAL for a subscriber. A stale slot fills the disk.
Replay	8	Resetting offsets and reading a Kafka topic again from the start. Possible because reading does not remove.
RLS	9	Row level security. A policy limiting which rows a role can see, enforced by the engine.
Rollback	1	Undoing everything a transaction did.
RPO	9	Recovery point objective. How much data you can afford to lose, measured in time.
RTO	9	Recovery time objective. How long you can afford to be down.

S

Term	Module	What it means
Saga	9	A long business process built from several transactions, undone by compensating transactions rather than a rollback.
SARGable	5, 10	A predicate an index can be used for. Wrapping a column in a function destroys this.
SCOPE_IDENTITY()	10	Returns the identity value generated in the current scope. @@IDENTITY can return a trigger's value instead.
SCHEMABINDING	10	Ties a SQL Server view or function to the tables it uses, so they cannot change underneath it. Required for an indexed view.
SECURITY DEFINER	9	Makes a procedure run with its owner's privileges. Must always be paired with a fixed <code>search_path</code> .
Selectivity	5	The share of rows a predicate matches. Low selectivity means an index is worth using.
Sequence	11	A generator of increasing numbers. Forgetting to reset one after a migration causes duplicate key errors on the first insert.
Serializable	6	The strictest isolation level. Transactions behave as if run one after another, at the cost of aborts.
Session (Redis)	7	A short-lived key holding a signed-in user's state, expired by TTL.
SKIP LOCKED	6	Makes a locking <code>SELECT</code> step over rows another transaction holds. How two relays share one outbox safely.
Split brain	9	Two nodes both believing they are primary, each accepting writes. Worse than an outage, because data diverges.
SSIS	10	SQL Server Integration Services. Microsoft's ETL tool.
SSMA	11	SQL Server Migration Assistant. Converts an Oracle schema and code to SQL Server.
Staging table	10, 11	A landing table with text columns and no constraints, so bad data can be loaded and then inspected.
Statistics	5, 10	The engine's summary of your data distribution. Stale statistics produce bad plans. A fresh migration has none.
Stored procedure	5	Named SQL stored in the database, invoked with <code>CALL</code> or <code>EXEC</code> . Can commit and return nothing.
Super key	4	Any set of columns that identifies a row uniquely, including sets with unnecessary extra columns.
Surrogate key	4	A meaningless generated identifier used as the primary key, in place of a natural one such as email.

T

Term	Module	What it means
tempdb	10	The SQL Server scratch database used by sorts, temp tables and version stores. Shared by every database on the instance.

Term	Module	What it means
THROW	10	The modern SQL Server statement for raising an error. Preserves the original error number.
Topic	8	A named stream of Kafka events, divided into partitions.
Transaction	1	A group of statements that succeed or fail as one unit.
Trigger	10	Code the engine runs automatically on insert, update or delete. Must be written set-based, not row-by-row.
TRUNCATE	2	Empties a table quickly. No WHERE, no triggers, and minimal logging.
TTL	7	Time to live. How long a cached key survives. Add jitter so keys do not all expire together.

U

Term	Module	What it means
UNION and UNION ALL	3	Stack two result sets. UNION removes duplicates and costs a sort. UNION ALL does not.
UNIQUE constraint	2	Guarantees no two rows share a value. Unlike a primary key, it may allow NULLs.
UPI	1	Unified Payments Interface. The Indian instant payments rail, and the highest-volume channel in the seed data.

V

Term	Module	What it means
Vacuum	6	The PostgreSQL process reclaiming space from old row versions. Blocked by long-running idle transactions.
View	5	A stored query that behaves like a table. Computed fresh on every read.

W

Term	Module	What it means
WAL	1	Write-ahead log. Changes are written to the log before the data pages, which is what makes durability possible.
Watermark load	11	Picking up changes since a stored timestamp. Misses deletes, and misses updates that do not touch the watermark column.
Window function	6	A function computing across related rows while keeping every row, using OVER. Running balances and rankings.
Write-behind	7	Writing to the cache and to the database later. Fast, and unusable for balances, because a crash loses money.
Write-through	7	Writing to the cache and the database together. The cache is never stale, and every write is slower.

The twenty terms worth knowing cold

If you remember nothing else from this sheet, remember these. Every one of them appears in interviews, and every one of them appears in more than one module of this programme.

Term	The one-line version
ACID	The four guarantees that make a database safe for money
Primary key	The column that identifies a row uniquely and never repeats
Foreign key	The pointer the database refuses to let you break
Normalization	Store each fact exactly once
Index	The structure that turns reading a million rows into reading four pages
Execution plan	The engine telling you what it is about to do, before you tune anything
SARGable	A predicate an index can actually be used for
Transaction	All of it happens, or none of it does
Isolation level	How much of other people's uncommitted work you can see
Lost update	Two reads, two writes, one change silently gone
Deadlock	Each waiting for what the other holds, fixed by consistent lock ordering
MVCC	Readers never block writers, and it costs you vacuum
Window function	Compute across rows without collapsing them
Cache-aside	Check the cache, miss, read the database, store the result
Invalidation	Delete the key, and do it after the commit
TTL	The backstop for the invalidation you forgot to write
Partition key	What decides ordering in Kafka, and there is no ordering across partitions
Idempotent	Safe to do twice, which is what makes at-least-once delivery acceptable
Outbox	Money and event in one transaction, published afterwards by a relay
RPO and RTO	How much you may lose, and how long you may be down